

Roll No.

--	--	--	--	--	--	--	--

End Semester Examination 2021

Name of the Course: B.Tech (ECE)

Semester: III

Name of the Paper: Electronic Devices and Circuits

Paper Code: **TEC 301**

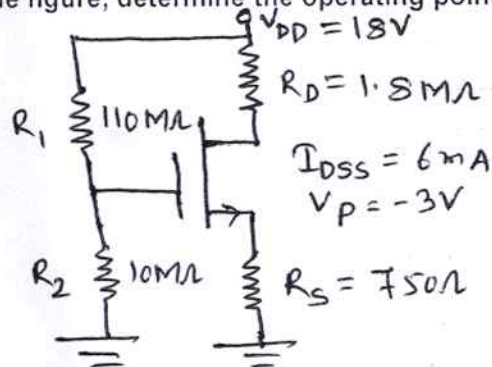
Time: 3 Hours

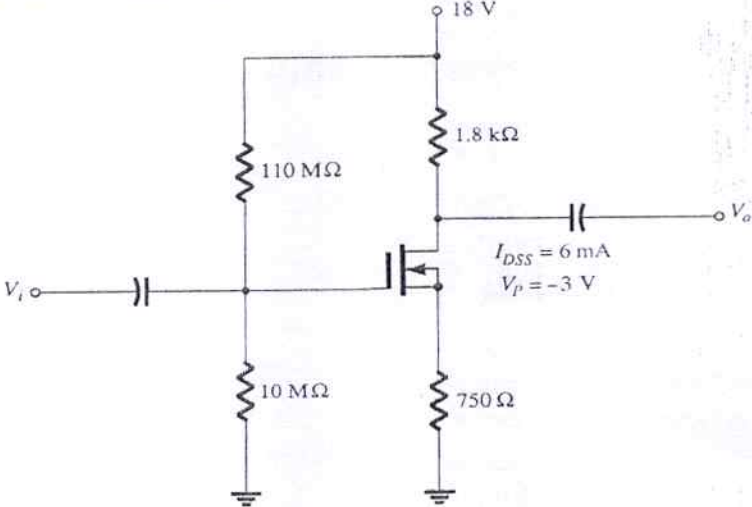
Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub-questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Discuss the input and output characteristics of common base configuration with a suitable diagram.	CO1, CO2
(b)	What is a multistage amplifier? Discuss the Darlington pair amplifier and find the formula for overall current gain.	
(c)	(i) The emitter current I_E in a transistor is 4 mA. If the leakage current I_{CBO} is 6 μA and $\alpha = 0.98$, find the collector and base currents. (ii) In a transistor, 99.6% of the carriers injected in to the base cross the collector-base junction. If the leakage current is 5 μA and the collector current is 20 mA. Calculate (i) the value of α_{dc} (ii) the emitter current.	
Q2	(20 marks)	
(a)	Discuss the output and transfer characteristics of n-channel depletion MOSFET with a suitable diagram.	CO1, CO2
(b)	For the network shown in the figure, determine the operating point using a graphical approach.	
(c)	Explain the following terms with respect to BJT (i) Early effect (ii) Punch Through effect	
Q3	(20 marks)	
(a)	Explain MOSFET as an amplifier, discuss its small signal model, and find the expression for transconductance (g_m).	CO3, CO5
(b)	Draw the a.c equivalent circuit of the self-bias configuration of N-type depletion MOSFET and find the value of voltage gain.	
(c)	What is feedback? Draw the general structure of feedback circuit and find the expression for overall gain in the case of negative feedback circuit.	
Q4	(20 marks)	
(a)	Explain with a neat circuit diagram, the working of Wein-Bridge oscillator. Derive an expression for frequency of oscillation.	CO3, CO4
(b)	Draw the circuit of the Colpitt oscillator and explain its working. Derive an expression for frequency of oscillation.	



(c)	Determine I_{DQ}, V_{GSQ} and V_{DS} for the given circuit. 	
Q5	(20 marks)	
(a)	Describe the working of transformer coupled Class A power amplifier in detail with the relevant diagram.	CO5, CO6
(b)	Describe the working of the Class B push pull power amplifier.	
(c)	Write the difference between a voltage amplifier and power amplifier and discuss the reason why we can't use voltage amplifier as a power amplifier.	